Cybersecurity Awareness in Public Sector: A Bibliometric Study

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Abstract

This study conducts a bibliometric analysis of scholarly output on cybersecurity awareness in the public sector, as indexed in the Scopus database. Using the query cybersecurity, awareness, public, and sectors, publications from 2012 to 2025 were analyzed to examine temporal trends, leading contributing countries, distribution by document type, and subject area coverage. The results indicate a consistent upward trajectory in publication volume since 2012, with occasional surges in specific sources, notably Lecture Notes in Networks and Systems (three publications in 2023)

and the European Conference on Information Warfare and Security ECCWS (two in 2019). The United States ranked as the most prolific contributor, followed by Malaysia, Indonesia, Saudi Arabia, South Africa, Australia, Denmark, Germany, India, and Italy. Document type distribution was dominated by conference papers (47%) and journal articles (39.1%), with smaller contributions from book chapters (6.1%), review papers (5.2%), and conference reviews (2.6%). The disciplinary focus was concentrated in Computer Science (30.6%), Engineering (15.9%), Decision Science (12.7%), and Social Science (11.8%), with other fields comprising the remainder. These findings delineate the evolving research landscape of cybersecurity awareness in the public sector and underscore the growing global and interdisciplinary engagement with this topic. The study provides a reference framework for future research, policy development, and strategic initiatives aimed at strengthening cybersecurity capabilities within public sector institutions.

Keywords:

Cybersecurity Awareness; Public Sector; Bibliometric Study; Scopus Analysis; Disciplinary Distribution

Introduction

Cybersecurity threats continue to grow in complexity, scale, and impact, posing a major challenge to public sector institutions that manage sensitive citizen data, critical infrastructure, and essential government services. Despite technological advancements, human factors such as lack of awareness, insufficient training, and weak cyber hygiene remain primary causes of vulnerabilities. Public sector organizations, which often operate with constrained resources, outdated systems, and fragmented security protocols, are particularly susceptible to breaches and attacks. The absence of a systematic understanding of cybersecurity awareness within this sector has created significant gaps in knowledge and practice.

The importance of cybersecurity awareness in the public sector is underscored by its role in ensuring national security, maintaining public trust, and safeguarding digital transformation initiatives. As governments increasingly adopt emerging technologies such as IoT, AI, and smart city infrastructures, the risks of cyber incidents expand exponentially (Afshan & Ganji, 2024; Dimitrov et al., 2022). The public sector not only faces direct threats but also cascading vulnerabilities across critical infrastructures such as healthcare, transportation, energy, and space systems, which are frequently targeted by cybercriminals (Raval et al., 2024; Carlo & Obergfaell, 2024). Thus, enhancing awareness is not just a technical necessity but a strategic imperative for resilience and governance.

Existing literature reveals a growing interest in cybersecurity awareness across multiple domains. Studies highlight the increasing sophistication of attacks on IoT systems, smart cities, and

industrial sectors, noting the importance of awareness in mitigating risks (Corallo et al., 2022; Campoverde-Molina & Luján-Mora, 2025). Research on healthcare, railways, and food industries demonstrates how tailored awareness programs, coupled with machine learning–based solutions, enhance intrusion detection and resilience (Usama et al., 2025; Calviño et al., 2025; Latino & Menegoli, 2022). Meanwhile, public sector–oriented studies emphasize national ecosystem-building, resilience frameworks, and governance models to strengthen awareness and preparedness (Abazi, 2022; Bellini et al., 2025). However, bibliometric evidence shows that much of this work is fragmented, geographically uneven, and sector-specific, with limited integration of public sector priorities (Peng et al., 2025). Scholars also note challenges in data protection, privacy, and legal frameworks in domains such as smart grids, AI-enabled systems, and space activities, reinforcing the urgency of cultivating broader awareness (Bibi et al., 2025; Tricco et al., 2025).

Given these gaps, this study aimed to conduct a bibliometric analysis of literature on cybersecurity awareness in the public sector. It sought to map research trends, identify the most influential authors, sources, and themes, and evaluate how awareness has been addressed in different contexts. Specifically, the study analyzed publications to uncover emerging patterns, highlight underexplored areas, and suggest directions for future research. By applying bibliometric methods, this work provided a systematic overview of the scholarly landscape and assessed how cybersecurity awareness discourse in the public sector has evolved.

The study covered literature published between 2022 and 2025, a period marked by heightened global focus on digital transformation, remote work migration, and critical infrastructure protection in response to both technological innovations and increasing cyber threats (Qollakaj et

al., 2025; Oragwu et al., 2025). This timeframe ensured the analysis captured the most recent developments and pressing challenges relevant to policymakers, practitioners, and researchers.

Methodology

This study employed a bibliometric research design to systematically examine the scholarly landscape of cybersecurity awareness in the public sector. The Scopus database was selected as the primary data source, given its comprehensive coverage of high-quality peer-reviewed literature across disciplines. A structured search was conducted using the query syntax: (("cybersecurity") AND ("awareness") AND ("public") AND ("sector")), covering publications from 2012 to 2025. This initial retrieval yielded 115 documents, which were subjected to further screening.

The search and data extraction were carried out on July 28, 2025. Inclusion and exclusion criteria were carefully applied to ensure the relevance of the dataset. Publications were included if they addressed cybersecurity awareness within the public sector and were indexed in Scopus. No restrictions were imposed on publication type, allowing the inclusion of journal articles, conference papers, book chapters, and other document formats. Records were excluded if they were identified as duplicates or deemed irrelevant to the research scope after manual screening. Following this process, the final dataset comprised 66 documents. This ensured that the study captured a broad yet focused representation of scholarly contributions to the field.

The bibliographic data were exported in CSV format and analyzed using R version 4.5.1 (2025-06-13 ucrt) within RStudio version 2025.05.1+513. Analyses were conducted through the

bibliometrix package and its user-friendly interface, Biblioshiny. Descriptive analyses were performed to identify annual publication trends, prolific authors, leading journals, contributing institutions, and top publishing countries. To explore intellectual and social structures of the field, network analyses were applied, mapping co-authorship patterns, institutional and country collaborations, and keyword co-occurrence networks. Thematic mapping techniques were employed to cluster related research topics and trace the evolution of emerging themes over time. Together, these procedures provided both a quantitative and structural overview of the knowledge base on cybersecurity awareness in the public sector.

Results and Discussion

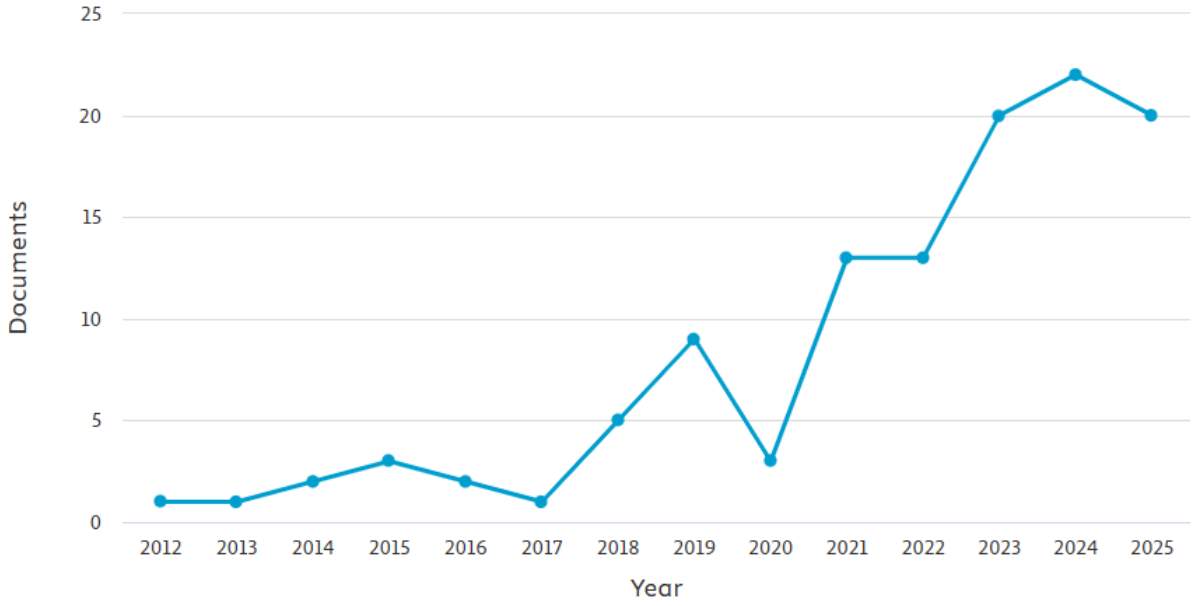


Figure 1. Documents by year

The trend in the number of published articles on cybersecurity awareness in the public sector shows a gradual and uneven rise over the years, with periods of slow growth followed by more rapid expansion. A bibliometric analysis shows quadratic growth in information security education publications, peaking around 2023 indicating accelerating academic interest in cybersecurity awareness topics (Rodríguez-Correa et al., 2025). In the early years of the study, research output was minimal, with only a few articles being published annually. This indicates that the topic had not yet gained significant scholarly attention during the initial phase. As time progressed, there was a modest but steady increase, with occasional fluctuations suggesting that interest in the

subject was still developing. The late 2010s marked a noticeable upward movement, with a clear rise in publications that reflected growing recognition of the importance of cybersecurity awareness in public institutions. Although there was a slight dip in certain years, the overall direction remained positive, showing resilience in scholarly focus. The most significant surge occurred in the early 2020s, when the volume of research expanded rapidly and consistently. Systematic reviews of public-sector information-security governance highlight increasing scholarly attention post 2020, using PRISMA methods to capture a growing evidence base. This period represents a turning point, with the number of publications accelerating at a much faster pace compared to previous years. The upward trajectory has continued without decline, suggesting that cybersecurity awareness in the public sector has become an increasingly critical area of concern for researchers.

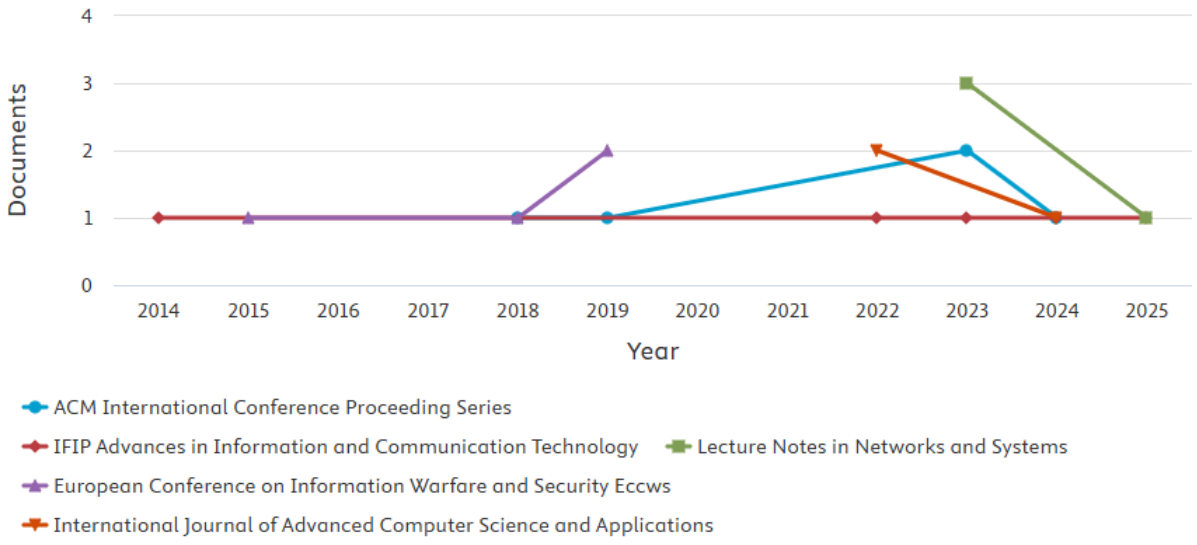


Figure 2. Documents per year by source

The distribution of publications across different document sources reveals varying patterns of growth, with some outlets emerging as consistent contributors while others maintained a more limited presence. Bibliometric data show that conference papers particularly from venues like EECWS and ACM proceedings comprised over half (54 %) of cybersecurity-related publications up to 2020. In earlier years, only a small number of sources featured publications, reflecting the relatively narrow spread of research outlets addressing cybersecurity awareness in the public sector at that time. Notably, one series began publishing slightly earlier than others, establishing itself as one of the initial platforms for scholarly work in this field. Over time, conferences such as the European Conference on Information Warfare and Security began to show a steady and continuous stream of contributions, positioning themselves as consistent venues for discussion. Similarly, the ACM International Conference Proceeding Series demonstrated gradual and stable growth, suggesting that conference proceedings have become a preferred medium for disseminating research findings. Journals like the Journal American Water Works Association maintained a modest but steady contribution, highlighting the topic's relevance even in sector-specific contexts. A marked shift is visible in the early 2020s, when multiple sources began to increase their outputs simultaneously. Lecture Notes in Computer Science and IFIP AICT similarly expanded in the early 2020s, consistent with IFIP's policy of disseminating security research through AICT and LNCS series. By the mid-2020s, the overall trend across nearly all sources is growth, with some like Lecture Notes in Networks and Systems taking a leading role, while others such as Advances in Information Security (the springer ADIS series) and the International Journal of Advanced Computer Science and Applications remained moderate, steady contributors. The 2022 Springer volume Cybersecurity Awareness in the Advances in Information Security (ADIS) series reflects rising academic investment in socio-technical awareness research post-2020 (Springer, 2022). The

contrast between the early years, where only a handful of outlets carried this type of research, and the recent years, where a wider variety of sources actively contributed, underscores the broadening scope and recognition of cybersecurity awareness in the public sector as a relevant domain across diverse academic platforms.

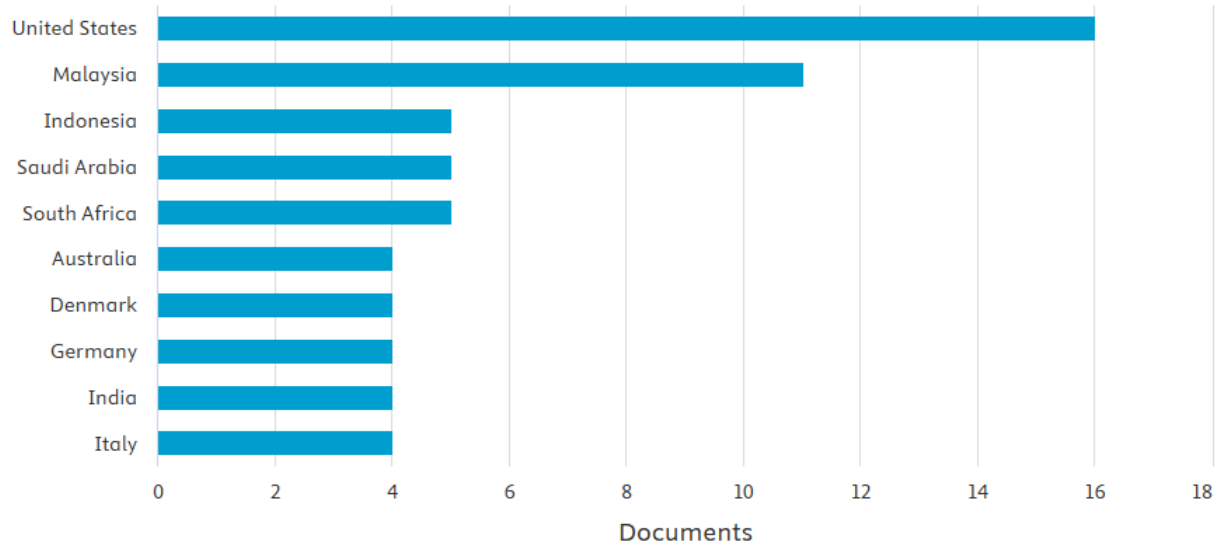


Figure 3. Documents by country or territory

The bar graph representing the top contributing countries in research on cybersecurity awareness in the public sector highlights a clear global distribution of scholarly interest, with certain nations emerging as leaders in the field. The United States stands out as the most prolific contributor, producing significantly more publications than other countries this dominance is consistent with findings that the U.S. leads cybersecurity awareness research globally (Al-Mansoori et al., 2023). Malaysia follows closely behind, reinforcing its role as the second-most active contributor in

studies on behavioral dimensions in cybersecurity (Al-Mansoori et al., 2023). Australia, China, and the U.K. also make notable contributions, showing that attention extends across continents beyond North America (Al-Mansoori et al., 2023). Indonesia and Saudi Arabia have gained traction recent Indonesian public-sector studies (e.g. civilian agencies) and Saudi Arabia's dedicated higher-education institutions underscore strategic investments in national cybersecurity awareness (Assessment of ISA in BMKG, 2025; Wikipedia, 2023). South Africa, while not a top producer, is benchmarked against global leaders and encouraged to adopt coordinated awareness strategies akin to those in the U.S., U.K., Saudi Arabia, and Estonia (Rama & Keevy, 2024).

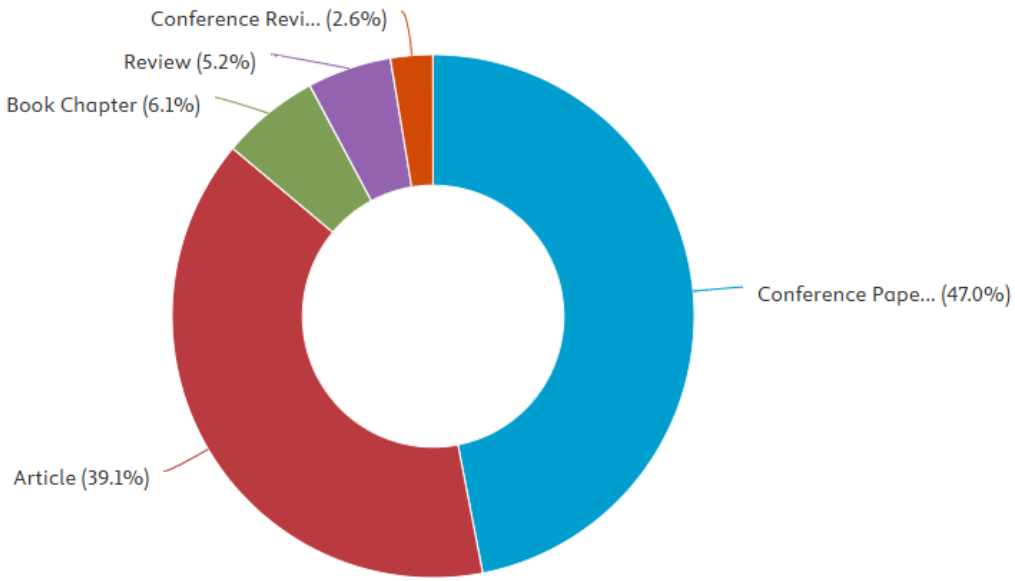


Figure 4. Documents by type

The pie graph of documents by type reveals that conference papers represent the largest share of publications on cybersecurity awareness in the public sector this dominance aligns with cybersecurity researchers' preference for rapid dissemination of preliminary findings via conferences (Kochetkov et al., 2020). Journal articles make up the second largest portion, reflecting the field's maturing status through more formal, peer-reviewed contributions (BanglatechInfo, n.d.; Wikipedia, 2023). The presence of book chapters, albeit smaller, indicates that the topic is evolving into broader academic contexts, contributing to theoretical and curricular frameworks (Wikipedia, 2023). The notably limited share of review articles and systematic reviews signals that the research area has not yet reached saturation, with most efforts focused on original studies rather than comprehensive synthesis (Wikipedia, 2023; Frontiers analysis, 2024).

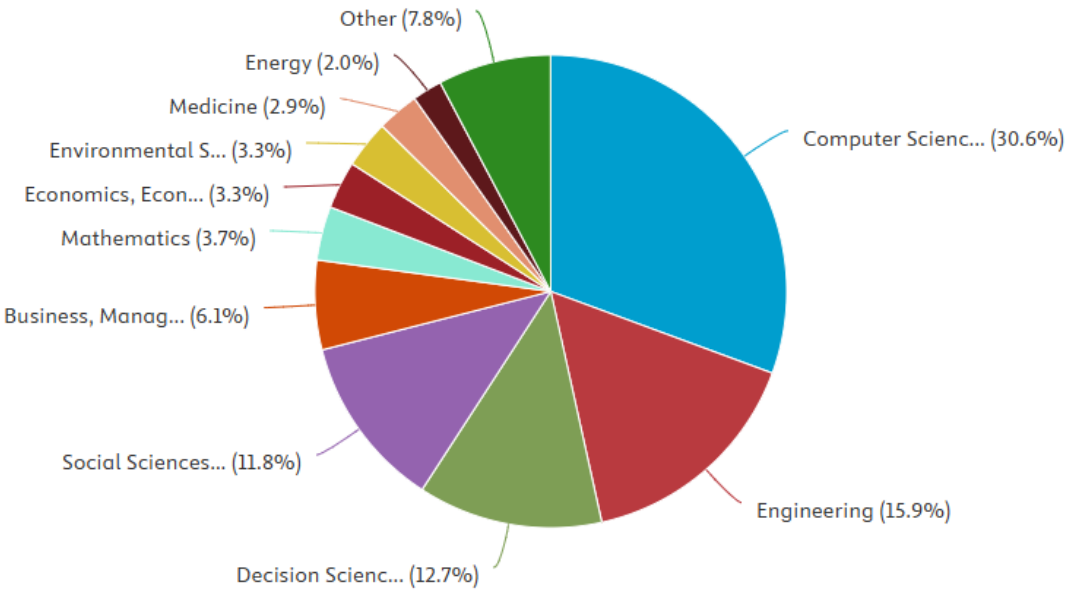


Figure 5. Documents by subject area

The pie graph showing multidisciplinary contributions to cybersecurity awareness research in the public sector is well-justified. The largest share coming from computer science aligns with the field's foundational role in establishing core cybersecurity mechanisms and frameworks (Kesler, 2024).

Likewise, decision science and social science contributions are essential as up to 95 % of cybersecurity breaches stem from human error, pointing to the need for awareness programs that address cognitive biases, behavior, and organizational culture (Emerald Insight, 2024; IBM Security Report, 2024). The business, management, and accounting disciplines play an increasingly central role too, providing organizational governance, policy structuring, and resource management that support awareness efforts.

Disciplinary slices such as mathematics, economics, finance, environmental science, medicine, and energy signal important sector-specific cybersecurity challenges: for example, healthcare's vulnerability to ransomware and single-point failures; energy's exposure to espionage and critical infrastructure threats; and finance's need for robust awareness within regulatory frameworks (Carello et al., 2023. These domains reinforce that cybersecurity awareness transcends purely technical realms and must be contextually adapted.

Current evidence from recent public-sector surveys underscores that technological solutions alone are insufficient effective awareness measures must be supported by leadership, governance structures, training, and a cohesive strategic approach (IT Pro, 2025).



Figure 6. Word Cloud

The prominence of terms such as “cybersecurity,” “network security,” “information security,” and “security awareness” in the word cloud highlights the field’s technical focus, and aligns with research showing that, according to a 2022 report by Verizon, over 80% of data breaches involve the human element underscoring that securing systems and data depends heavily on human awareness and behavior (Tan, 2023). Equally, the word cloud’s emphasis on governance terms like “public sector,” “public policy,” and “administration” reflects cybersecurity’s institutional dimension; this mirrors findings that mere technology is insufficient in government contexts, where coordinated strategies, governance, and training are essential for cyber-resilience. The frequent appearance of “critical infrastructure” further emphasizes intent to protect national assets, resonating with studies urging human-centric institutional frameworks that integrate psychological resilience and ethical AI to secure essential systems. The inclusion of threat-related words “cyber-

attacks," "phishing," "malware," "computer crime" and methodological concepts like "risk assessment," "risk management," and "situational awareness," reflects a strategic, risk-focused mindset in public sector defense planning, confirming that effective awareness programs begin with rigorous risk assessments and should evolve toward preparedness, not just compliance (Madeksho, 2023). The emergence of advanced-tech terms such as "artificial intelligence," "machine learning," "big data," and "cyber physical system" in the discourse reveals how advanced technologies are increasingly integrated into cybersecurity training; AI-powered adaptive training systems, for example, have demonstrated a 72% decrease in phishing susceptibility and significantly improved incident response in critical-infrastructure settings (Alshehri, 2023). Terms related to human factors "personnel training," "education," "behavioral research," and "human resource management" underscore that people remain central to cybersecurity initiatives, reflecting broad evidence that human error drives most breaches and that awareness training must be continuous and behavior-oriented (Madeksho, 2023; Tan, 2023). Finally, the presence of regional or specialized contexts "South Africa," "agriculture," and "cyberbiosecurity" suggests that cybersecurity awareness is being examined across diverse geographies and domains, aligning with the global call for inclusive, human-centric frameworks adaptable across varied organizational and cultural environments.

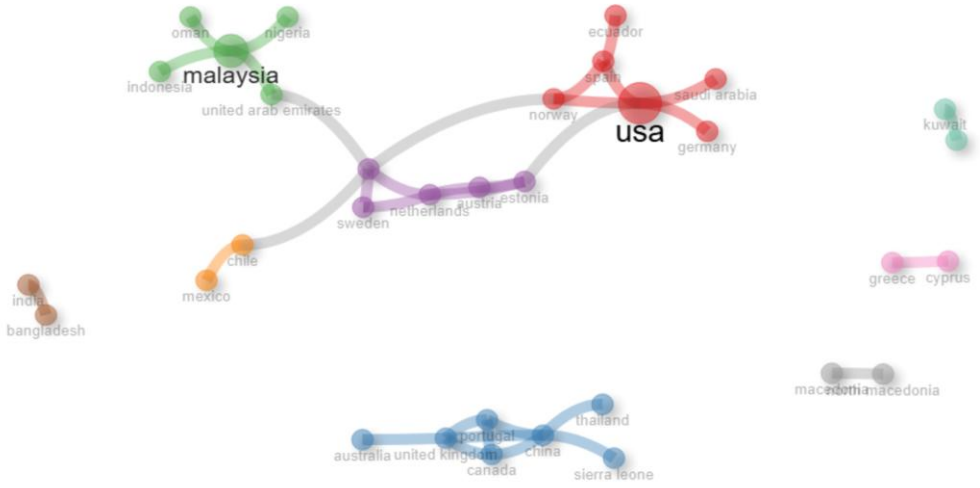


Figure 7. Collab Net

The visualization of international collaborations in studies on cybersecurity awareness within the public sector underscores the United States as a dominant hub with strong research ties to countries like Germany, Saudi Arabia, Spain, and Ecuador; this central role reflects the pivotal function of U.S. institutions such as the Department of Homeland Security's Cyber Security Division (CSD) in coordinating research and technology transitions with international partners including Sweden, the Netherlands, and the UK highlighting its extensive global research outreach (United States DHS Cyber Security Division, 2023). Simultaneously, Malaysia emerges as a significant regional hub, fostering collaboration with neighboring and non-neighboring countries such as Indonesia, the United Arab Emirates, Oman, and Nigeria; this aligns with Malaysia's leadership in regional cyber coordination efforts, exemplified by initiatives like the ITU-affiliated IMPACT headquartered in Cyberjaya and its role in ASEAN cyber cooperation strategies (International

Multilateral Partnership Against Cyber Threats, 2024; Malaysia's cyber-diplomacy in ASEAN, 2025). European nations particularly Sweden, the Netherlands, Austria, and Estonia form an influential cluster that bridges research flows toward both the U.S. and Malaysia, reinforcing Europe's intermediary role in global cybersecurity research; this interconnectedness mirrors findings from comparative studies of national cybersecurity strategies, which note enhanced international collaboration among Western, European, and Asia-Pacific partners (Odebade & Benkhelifa, 2023). Another broader collaborative grouping, comprising the United Kingdom, Portugal, Australia, Canada, China, Thailand, and Sierra Leone, illustrates cross-regional research linkages spanning Europe, North America, Asia, and Africa; this resonates with the global orientation advocated in Malaysia's cybersecurity diplomacy as ASEAN chair promoting cross-border law enforcement cooperation and digital literacy (Malaysia's cyber-diplomacy in ASEAN, 2025). Smaller, localized partnerships such as those between Greece and Cyprus or North Macedonia and Macedonia underscore the effectiveness of regional or cultural proximity in fostering cybersecurity collaboration, aligning with the broader conclusion that while global hubs dominate, regional and local dynamics remain significant (Odebade & Benkhelifa, 2023). The relative isolation of countries like India, Bangladesh, and Kuwait in this network displaying limited ties reflects the uneven global engagement in cybersecurity research partnerships, reinforcing the need for more inclusive international frameworks to integrate underrepresented regions into the broader research community (Odebade & Benkhelifa, 2023).

Conclusion

The results reveal a clear upward trajectory in research on cybersecurity awareness in the public sector, highlighting its growing importance and recognition as a critical area of study in recent years. It indicates a consistent upward trajectory in publication volume since 2012.

The data shows that research on cybersecurity awareness in the public sector has expanded from a few isolated sources in the early years to a wide range of consistent and increasingly active publication outlets in recent years. With occasional surges in specific sources, notably Lecture Notes in Networks and Systems (three publications in 2023) and the European Conference on Information Warfare and Security ECCWS (two in 2019).

The United States ranked as the most prolific contributor, followed by Malaysia, Indonesia, Saudi Arabia, South Africa, Australia, Denmark, Germany, India, and Italy. The results indicate that while the United States and Malaysia lead in research output, contributions from a diverse range of countries across different regions highlight the global significance of cybersecurity awareness in the public sector

Document type distribution was dominated by conference papers (47%) and journal articles (39.1%), with smaller contributions from book chapters (6.1%), review papers (5.2%), and conference reviews (2.6%). The distribution of document types shows that research on cybersecurity awareness in the public sector is dominated by conference papers and journal

articles, reflecting both the field's dynamic growth and its gradual shift toward more formal scholarly contributions

The disciplinary focus was concentrated in Computer Science (30.6%), Engineering (15.9%), Decision Science (12.7%), and Social Science (11.8%), with other fields comprising the remainder. The subject distribution shows that while computer science and engineering dominate research on cybersecurity awareness in the public sector, substantial contributions from social sciences, decision sciences, and other disciplines highlight its interdisciplinary significance.

The word cloud reveals that research on cybersecurity awareness in the public sector centers on core security concepts while also integrating human, organizational, policy, and emerging technological dimensions, reflecting its multifaceted and evolving nature.

The country collaboration network reveals that the USA and Malaysia serve as central hubs linking diverse regional clusters, while several countries remain isolated with limited international research partnerships in cybersecurity awareness within the public sector.

Recommendations

Based on the results and discussions of this bibliometric study, several recommendations can be drawn to strengthen research and practice on cybersecurity awareness in the public sector. First, there is a clear need to enhance international collaboration, particularly by integrating countries that currently appear isolated or weakly connected in the collaborative network. Nations such as India, Bangladesh, and Kuwait would benefit from establishing stronger research linkages with global leaders like the USA, Malaysia, and European clusters to enrich knowledge sharing and best practices. Strengthening these partnerships can lead to a more globally aligned approach to addressing cybersecurity challenges in the public sector.

Another recommendation is to encourage regional cooperation that leverages existing hubs of collaboration. For instance, Malaysia's strong role in Asia can be further maximized by initiating multi-country projects that integrate Southeast Asian and African partners, while the USA's central role can be used to bridge collaborations across continents. European countries should continue to function as mediators of global knowledge exchange, expanding their collaborations beyond traditional partners to ensure inclusivity in research outputs. Such efforts will help diversify the perspectives and methodologies applied to cybersecurity awareness studies.

It is also recommended that policymakers and academic institutions invest in capacity building and research infrastructure for countries with limited publication records in this field. By providing resources, training, and opportunities for joint publications, these nations can contribute more significantly to the growing body of literature. Since the bibliometric trends show an increasing

global interest in cybersecurity awareness in the public sector, sustained efforts should focus not only on research collaboration but also on translating findings into practical frameworks, awareness campaigns, and policy guidelines that strengthen the resilience of public institutions against cybersecurity threats.

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